

Reducibility

language

— Reduction: convert problem A to B s.t the solution to B can be used to solve A .

$$(A \leq B, A \alpha B)$$

Formal definition is as follows.

Language A is mapping reducible to language B , written as $A \leq_m B$, if there is a computable function $f: \Sigma^* \rightarrow \Sigma^*$, where for every w ,

$$w \in A \iff f(w) \in B$$

A computable function is basically an algorithm that takes an input and output something in finite amount of time.

How can we use this to show
the undecidability?

Right now we proved A_{TM} problem is undecidable.

We are going to prove $HALT_{TM}$ is undecidable.

$$HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on } w \}$$

$HALT_{TM}$ and A_{TM} are two different languages.

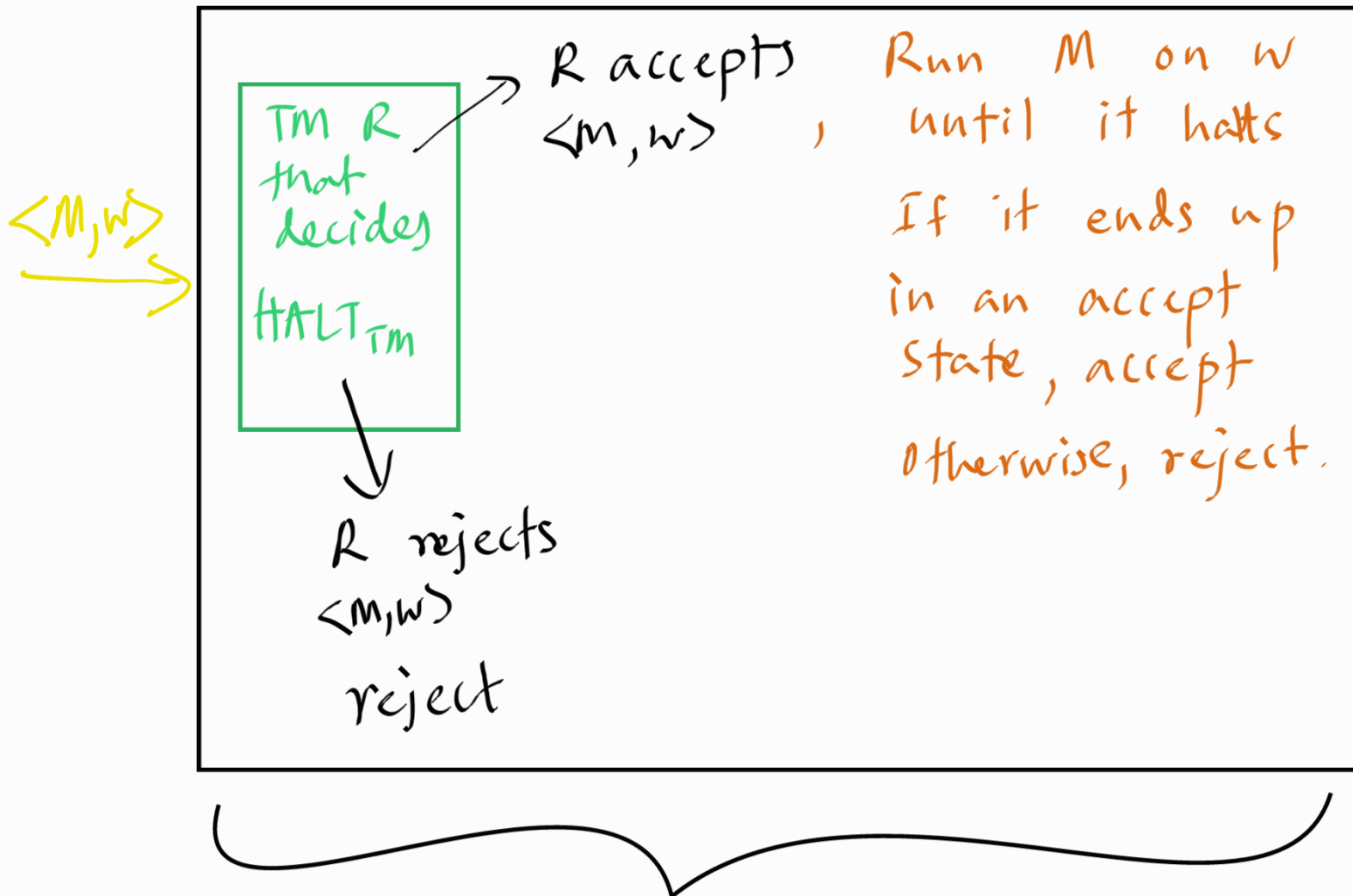
We are going to prove $HALT_{TM}$ is undecidable.

We will show that

$$A_{TM} \leq HALT_{TM}$$

(Note that order of the reduction matters)

Assume HALT_{TM} is decidable, let TM R be the decider for HALT_{TM} .



A decider for A_{TM}

But A_{TM} is undecidable, therefore the decider that we created cannot exist. That means our initial assumption is incorrect.

$\therefore \text{HALT}_{\text{TM}}$ is undecidable.

Let's write the proper proof!

Theorem 5.1

$HALT_{TM}$ is undecidable

Proof Idea: We will construct a decider for A_{TM} (which we know that is undecidable) using a decider for $HALT_{TM}$.

Assume $HALT_{TM}$ is decidable, and the turing machine R is that decider.

Let us construct Turing machine S that decides A_{TM} .

$S =$ on input $\langle m, w \rangle$ where M is a TM and w is a string

1. Run TM R on $\langle m, w \rangle$
2. If R rejects $\langle m, w \rangle$
reject
3. If R accepts, simulate M on w until it halts.
4. If M halts on an accept state, accept
If M halts on a reject state, reject.

Clearly TM S is a decider for A_{TM} . But A_{TM} is undecidable. Therefore, our initial assumption must be false.

$HALT_{TM}$ is undecidable.

$$E_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$$

Theorem 5.2

E_{TM} is undecidable.

E_{TM} is basically the problem where we check a TM accepts nothing.

I am going to reduce A_{TM} to E_{TM}

$$A_{TM} \leq E_{TM}$$

$$\langle M, w \rangle \qquad \langle M, \rangle$$

Assume E_{TM} is decidable and
TM R decides it.

I will attempt to create TMs
that decides A_{TM} .

first we will look at how we
can create the following special
TM M_1 , by encoding the input $\langle m, w \rangle$

$M_1 =$ "on input $\langle x \rangle$, where x is a
string and $\langle m, w \rangle$ is hard coded
inside of this machine.

1. If $x \neq w$, reject

2. If $x = w$

run M on w and
accept x , if M accepts w .

M accepts $w \Rightarrow L(M_1) = \{w\}$

M does not
accept $w \Rightarrow L(M_1) = \emptyset$

$S =$ "on input $\langle m, w \rangle$, where M is a TM and w is a string

1. Construct special TM M_1 that encodes $\langle m, w \rangle$ inside it.
2. Run R on $\langle M_1 \rangle$
3. If R accepts, **reject**
If R rejects, **accept**.

Note that S is a decider for A_{TM} , but A_{TM} is undecidable

$\therefore E_{TM}$ is undecidable.

$FINITE_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is finite} \}$

$$A_{TM} \leq FINITE_{TM}$$

$\langle m, w \rangle$ $\langle M \rangle$

Assume $FINITE_{TM}$ is decidable, and TM R decides it.

Now we construct TM S that is a decider for A_{TM} .

$M_1 =$ on input $\langle x \rangle$, hard code $\langle m, w \rangle$ inside it

1. Run M on w , accept x if M accepts w .

$$M \text{ accept } w \Rightarrow L(M_1) = \Sigma^*$$

$$M \text{ does not accept } w \Rightarrow L(M_1) = \emptyset$$

$S =$ on input $\langle M, w \rangle$, where M is a TM and w is a string

1. Construct a special TM M_1 that encodes $\langle M, w \rangle$ in its construction.
2. Run R on $\langle M_1 \rangle$
3. If R accepts, reject
4. If R rejects, accept

S is a decider for A_{TM} ,
but A_{TM} is undecidable.

$\therefore$ our initial assumption is incorrect.
 $\therefore$ $FINITE_{TM}$ is undecidable.

2-DIFF-TM = $\{ \langle M \rangle \mid M \text{ is a TM and } L(M) \text{ contains two different strings of same length} \}$

$A_{TM} \leq 2\text{-DIFF-TM}$

$\langle M, w \rangle$ $\langle M, \rangle$

construct special TM M_1 that encodes $\langle M, w \rangle$

$M_1 = "$ on input $\langle x \rangle$, where it encodes $\langle M, w \rangle$

1. If $x = aa$ or $x = bb$

Run M on w and accept x
if M accepts w .

2. Else

reject.

$M \text{ accepts } w \Rightarrow L(M_1) = \{ aa, bb \}$

$M \text{ does not accept } w \Rightarrow L(M_1) = \emptyset$

Assume 2-DIFF-TM is decidable
and TM R decides it.

Then construct TM S that decides
 A_{TM} as follows:

$S =$ "on input $\langle m, w \rangle$, where M is a TM
and w is a string

1. construct M_1 that encodes $\langle m, w \rangle$
as described earlier.

2. Run R on $\langle M_1 \rangle$

3. If R accepts; accepts

If R rejects, reject.

clearly S is a decider for A_{TM} .

This is a contradiction.

$\therefore$ 2-DIFF-TM is undecidable.

